# Operations Research 1: Deterministic Models
# LP Modeling

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# Wyndor Example

# Wyndor Example

- Wyndor Glass Co.
  - Two products
    - Product 1: an 8-foot glass door
    - Product 2: a 4x6 foot wood-frame window
  - Three plants
  - Limited production time (per week)
  - “Infinite” demands

| | Doors | Windows | | | |
|---|---|---|---|---|---|
| Profit per batch | $3,000 | $5,000 | | | |
| | | | | | Hours |
| | Hours used per batch produced | | | | available |
| Plant 1 | 1 | 0 | | | 4 |
| Plant 2 | 0 | 2 | | | 12 |
| Plant 3 | 3 | 2 | | | 18 |
| | | | | | |
| | Doors | Windows | | | Total profit |
| Batches produced | | | | | |

# Wyndor Example

- $x_1$ = batches of product 1 per week
- $x_2$ = batches of product 2 per week
- $Z$ = total profit per week

| | Doors | Windows | | |
|---|---|---|---|---|
| Profit per batch | $3,000 | $5,000 | | |
| | | | | Hours |
| | Hours used per batch produced | | | available |
| Plant 1 | 1 | 0 | | 4 |
| Plant 2 | 0 | 2 | | 12 |
| Plant 3 | 3 | 2 | | 18 |

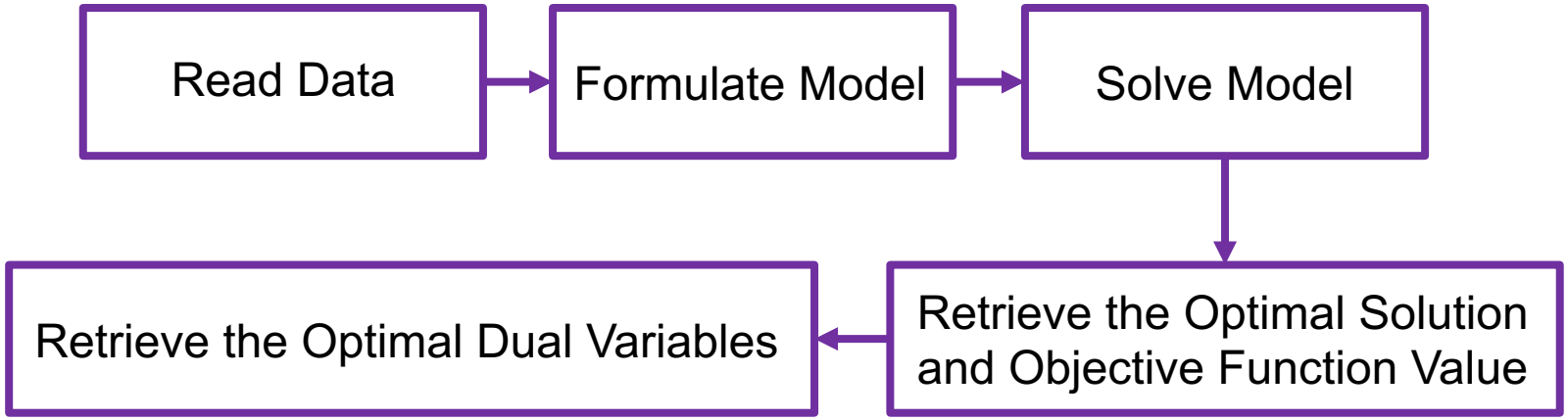
$$
\begin{array}{lrrl}
\max & Z = & 3x_1 \quad +5x_2 & \\
\text{s.t.} & & x_1 & \leq 4, \\
& & 2x_2 & \leq 12, \\
& & 3x_1 \quad +2x_2 & \leq 18, \\
& & x_1, \quad x_2 & \geq 0.
\end{array}
$$

# An Exemplary Program

- Notes
  - This exemplary program focuses on illustrating how to formulate and optimize an LP model using Gurobi in Python.
  - We omit the I/O parts, which are elaborated in the tutorial *I/O Between Python and Excel*.

# An Exemplary Program

- main.py
  - First, read the Dictionary objects that contain the input data.
  - Next, formulate a Gurobi model instance.
    - Using the Dictionary objects.
  - Then solve the Gurobi model instance, retrieve the optimal solution, and the objective function value.
  - Finally, retrieve the optimal dual variables.



# An Exemplary Program

- model.py
  - Formulate and solve the Wyndor example using Gurobi.
  - Functions
    - formulateModel(): Formulate a Gurobi model instance based on given Dictionary objects.
    - solveModel(): Optimize the given Gurobi model instance.
    - getOptimalDualVariableValues(): Get optimal dual variable values of the corresponding constraints.
    - writeModel(): Save the given Gurobi model to a file, either in MPS or LP format.
    - readModel(): Load a Gurobi model from a MPS or LP file.

# Basic Functions in Gurobi 12.0

# Basic Functions in Gurobi 12.0

- Summary of Basic Python Functions in Gurobi 12.0

| Description | Python Function |
|---|---|
| Create Gurobi model | gurobipy.Model() |
| Add variables | Model.addVar() |
| | Model.addVars() |
| Set objective function | Model.setObjective() |
| Add constraints | Model.addConstr() |
| | Model.addConstrs() |
| Optimize Gurobi model | Model.optimize() |
| Retrieve variables | Model.VarByName() |
| | Model.Vars() |
| Retrieve linear constraints | Model.getConstrByName() |
| | Model.getConstrs() |

# Basic Functions in Gurobi 12.0

- Create a Gurobi model.
  - Function gurobipy.Model()
    - Details & examples: https://docs.gurobi.com/projects/optimizer/en/12.0/reference/python/model.html#Model.Model

# Basic Functions in Gurobi 12.0

- Add variables to a Gurobi model.
  - Function Model.addVar()
    - Add ONE decision variable to a Gurobi model.
    - Details & examples: https://docs.gurobi.com/projects/optimizer/en/12.0/reference/python/model.html#Model.addVar
  - Function Model.addVars()
    - Add MULTIPLE decision variables to a Gurobi model.
    - Details & examples: https://docs.gurobi.com/projects/optimizer/en/12.0/reference/python/model.html#Model.addVars

# Basic Functions in Gurobi 12.0

- Set the objective function of a Gurobi model.
  - Function Model.setObjective()
    - Details & examples: https://docs.gurobi.com/projects/optimizer/en/12.0/reference/python/model.html#Model.setObjective

# Basic Functions in Gurobi 12.0

- Add constraints to a Gurobi model.
  - Function Model.addConstr()
    - Add ONE constraint to a Gurobi model.
    - Details & examples: https://docs.gurobi.com/projects/optimizer/en/12.0/reference/python/model.html#Model.addConstr
  - Function Model.addConstrs()
    - Add MULTIPLE constraints to a Gurobi model.
    - Details & examples: https://docs.gurobi.com/projects/optimizer/en/12.0/reference/python/model.html#Model.addConstrs

# Basic Functions in Gurobi 12.0

- Optimize a Gurobi model.
  - Function Model.optimize()
    - Details & examples: https://docs.gurobi.com/projects/optimizer/en/12.0/reference/python/model.html#Model.optimize

# Basic Functions in Gurobi 12.0

- Retrieve variables from a Gurobi model.
  - Function Model.VarByName()
    - Retrieve ONE variable from its name. If multiple variables have the same name, this method chooses one arbitrarily.
    - Details & examples: https://docs.gurobi.com/projects/optimizer/en/12.0/reference/python/model.html#Model.getVarByName
  - Function Model.Vars()
    - Retrieve a list of ALL variables in the model.
    - Details & examples: https://docs.gurobi.com/projects/optimizer/en/12.0/reference/python/model.html#Model.getVars

# Basic Functions in Gurobi 12.0

- Extract the dual variable of a constraint.
  - The dual variable corresponding to a constraint is extracted by obtaining the "Pi" attribute of the constraint after optimizing the model.
  - Function Model.getConstrByName()
    - Retrieve a linear constraint from its name.
    - Details & examples: https://docs.gurobi.com/projects/optimizer/en/12.0/reference/python/model.html#Model.getConstrByName
  - Function Model.getConstrs()
    - Retrieve a list of all linear constraints in the model and return all linear constraints in the model.
    - Details & examples: https://docs.gurobi.com/projects/optimizer/en/12.0/reference/python/model.html#Model.getConstrs

# Further Information

- Gurobi official tutorials
  - https://support.gurobi.com/hc/en-us/articles/14799677517585-Getting-Started-with-Gurobi-Optimizer
  - https://docs.gurobi.com/current/index.html
  - https://www.gurobi.com/resource/level-2-beginner/
- Gurobi Python API overview
  - https://docs.gurobi.com/projects/optimizer/en/current/reference/python.html